AUM SRI SAIRAM

SUPPORT VECTOR MACHINE

Large Margin Classifier:

Questions:

1. What type of machine learning task is resolved using SVM?
2. What type of classification task is implemented using SVM?
3. What is the decision boundary in SVM and its mathematical/vector representation?
4. How do you visualize a hyperplane in a 3-dimensional space?
5. Why it is said that the weight vector is normal/orthogonal to the hyperplane, proof?
6. What are the classes defined in an SVM based on the hyperplane?
7. What is a convex hull?
8. Why do we call classification task using SVM as large margin Classifier?
9. What is the distance of a point to the hyperplane, and how to calculate it?
10. What is the distance of the origin to the hyperplane?
11. Why the distance calculated in Q8 is called signed distance?
12. How to convert the signed distance into unsigned distance?
13. What is margin and large margin?
14. What is the objective function of a large margin classifier, to realize it using necessary transformations?
15. What are the unknown parameters in the objective function?
16. How to solve a primal convex minimization problem with quadratic inequalities using optimization techniques?
17. What do you mean by dual, how do you realize it in SVM?
18. What are the values of unknown parameters, obtained it?
19. What is the final result of the SVM classifier for a new point?

SOFT MARGIN CLASSIFIER:

1. **What do you mean by soft margin? /Linearly -inseparable?**
2. **How to handle the misclassified points in SMC?**
3. **What is the objective function of SMC?**
4. **What is the meaning of regularization and Hinge loss?**
5. What are the unknown parameters in the objective function?
6. How to solve a primal convex minimization problem with quadratic inequalities using optimizations techniques?
7. What do you mean by dual, how do you realize it in SMC?
8. What are the values of unknown parameters, obtain it?
9. What is final result of SMC classifier for a new point?